COURSE GUIDE & SYLLABUS · GRADES 3–5 · AGES 8–10

Scratch (Gr 3–5) Beginner: Scratch Game Makers

Code your own animations and games — from your first script to your own arcade game.

A 12-lesson project-based coding course. Learners build real programs in the built-in Code Lab (or Scratch): an animated name, a dance party, a catching game and their own arcade game. Every lesson has an explainer video, a build-along, block-coding challenges that are checked automatically, a printable worksheet and a teacher lesson plan.

What learners will be able to do

- Write programs as sequences of instructions and explain why order matters
- Use events (green flag, keys, clicks, messages) to start scripts
- Use repeat and forever loops for patterns and animation
- Use if-then and if-else with sensing conditions to make decisions
- Use x–y coordinates and random numbers to position and move sprites
- Create and use variables such as score, lives and time
- Coordinate sprites with broadcast messages and build win/lose endings
- Plan, build, test, debug, improve and present an original game

How every lesson works

Watch	A short animated explainer video with captions and read-aloud narration.
Learn	Learn cards with the real coding blocks, and a step-by-step build-along.
Try it (Code Lab)	Code in the built-in Scratch-style Code Lab with a live stage. Challenges are checked automatically and give a clear next step; hints unlock one at a time.
Puzzles & quick checks	Robot puzzles and questions that build understanding, with instant feedback.
Probe	Debugging and “new situation” tasks to show real understanding.
Projects	Each module ends with a creative project built in the Code Lab and reviewed by the teacher with a rubric.
Printables	A worksheet for every lesson, a project brief for every project, and a lesson plan with answer keys for the teacher.

Scope and sequence

#	Lesson	Big idea	Learners build

1	**Meet Scratch: Sprites, the Stage and Your First Script** Find the stage, sprites, block palette and code area, and build a script that runs blocks in order (a sequence).	Algorithms and sequence	Your first script
2	**Events: Make Things Happen When…** Use event hat blocks (green flag, key pressed, sprite clicked) to start scripts, and control a sprite with the arrow keys.	Events	Drive your sprite with the arrow keys
3	**Project: My Animated Name** Plan, build, test and share an interactive name project that uses events and sequences, with a green-flag reset script.	Project: sequences + events	Project: My Animated Name
4	**Repeat Loops: Do It Again!** Spot repeating patterns and use a repeat loop to run blocks many times, including drawing shapes with the pen.	Loops (repeat)	Loops and the pen
5	**Forever Loops and Costume Animation** Use forever loops, costumes and wait blocks to animate a sprite, and make it walk and bounce off the edges.	Loops (forever) and animation	Make a sprite walk
6	**Project: Dance Party** Plan, build, test and share an animated dance party that uses loops, costumes, sound and backdrop changes.	Project: loops + animation	Project: Dance Party
7	**If-Then: Teaching Sprites to Decide** Use if-then and if-then-else blocks with sensing conditions (touching, key pressed) inside loops so sprites react to what happens.	Conditionals and sensing	Don’t touch the lava!
8	**Coordinates and Random: Things That Fall** Use x and y coordinates and “pick random” to place sprites, and make an object fall and restart from a random place at the top.	Coordinates and randomness	Make it rain apples
9	**Project: Catch the Apple** Combine events, loops, conditions, coordinates and random numbers to build and test a playable catching game.	Project: game play	Project: Catch the Apple
10	**Variables: Keeping Score** Create a variable, set it at the start, change it during the game, and use it in a condition.	Variables	Add a score to Catch the Apple
11	**Messages, Game Over and Debugging** Use broadcast messages to coordinate sprites, add win and game-over endings with a timer, and debug programs step by step.	Broadcasts, game states and debugging	Win and game-over screens

12	**Capstone: My Arcade Game** Design, build, playtest, improve and present an original game that uses events, loops, conditions, variables and messages.	Capstone: everything together	Project: My Arcade Game (capstone)

Modules

Module 1 · Hello, Scratch! Sequences and events
Meet the stage and sprites, write your first scripts, and make things happen when keys are pressed or sprites are clicked. Project: My Animated Name.

Big question: How do we tell a computer exactly what to do — and when?

Module 2 · Loops and animation
Spot patterns, use repeat and forever loops, draw with the pen and animate with costumes. Project: Dance Party.

Big question: How can a few blocks do a lot of work?

Module 3 · Decisions and game play
Teach sprites to decide with if-then, use coordinates and random numbers, and build your first real game. Project: Catch the Apple.

Big question: How do games react to what the player does?

Module 4 · Variables and game design
Keep score with variables, connect sprites with messages, debug like a pro, and design your own arcade game. Capstone: My Arcade Game.

Big question: How do we design a complete game that others want to play?

Assessment

Coding challenges are checked on the server by running the learner's program. Quick checks give instant feedback. Written answers and projects are reviewed by the teacher against 4-level rubrics (Beginning · Developing · Secure · Excellent).

What you need

- Laptop, Chromebook or tablet with a modern browser
- The built-in Kitabash Code Lab (no account needed)
- Optional: Scratch (scratch.mit.edu or the Scratch app)
- Printed worksheets (optional)

Before you start

- Basic mouse or touch skills
- Reading simple sentences (videos have captions and an optional read-aloud)

Safety and privacy

No personal information is requested. The Code Lab runs inside Kitabash; sharing to scratch.mit.edu is optional and needs a parent-approved account for under-13s. Learners under 18 use the AI tutor only with guardian consent.

What comes next

After this course: **Scratch (Gr 3–5) Intermediate**.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.